Introduction to Artificial Intelligence

1. What is Artificial Intelligence (AI)?

Imagine this…

👉 You ask your phone: *"Hey Google, what's the weather?"*
👉 It understands, thinks, and answers.

That's AI.

Simple Definition:

- Artificial Intelligence (AI) is a branch of computer science that focuses on developing machines or systems capable of performing tasks that normally require human intelligence.

In simple words:

AI means teaching computers to:

- Think
- Learn
- Make decisions
- Understand language

Real-Life Examples

- Google Assistant / Siri
- YouTube recommendations
- Self-driving cars
- ChatGPT

How AI Works (basic idea)

AI uses:

- Data
- Algorithms
- Learning

It studies patterns and improves over time.

Human VS AI (Easy understanding)

Human	AI
• Learns from experience.	• Learns from data.
• Has emotions.	• No emotions.
• Can think creatively.	• Follows logic.

Explanation:

AI aims to simulate human intelligence in machines so that they can think, act, and respond like humans in different situations. Instead of simply following predefined instructions, AI systems are designed to analyze data, identify patterns, and improve their performance over time.

Artificial Intelligence is widely used in various real-world applications. For example, virtual assistants such as Google Assistant and Siri use AI to understand voice commands and provide appropriate responses. Similarly, recommendation systems used by platforms like YouTube and Netflix analyze user behaviour to suggest relevant content. Self-driving cars use AI to recognize objects, make driving decisions, and ensure safety.

AI systems can perform tasks faster and more accurately than humans in certain areas. However, unlike humans, they do not possess emotions or true understanding; they rely entirely on data and programmed logic.

In summary, Artificial Intelligence refers to the simulation of human intelligence in machines that are programmed to think, learn, and act intelligently.

Why AI is Important

- Reduces human work
- Solves complex problems
- Works faster than humans
- Used in almost every field